

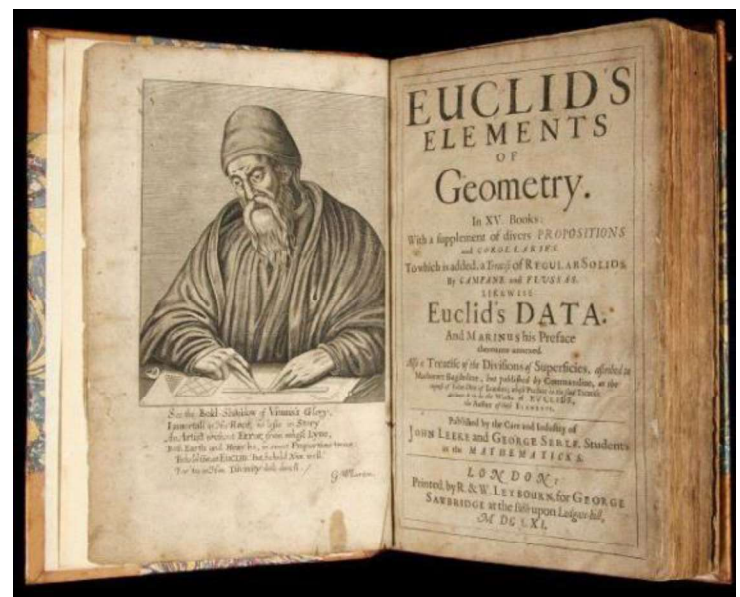
Geometry

Friday, May 17, 2024

11:44 AM

Euclid

- The Father of Geometry
- Lived 325 - 265 BC in Alexandria, Egypt
- Wrote "The Elements," his treatise of Geometry
- He defined many "axioms," i.e., statements considered true but are not proven

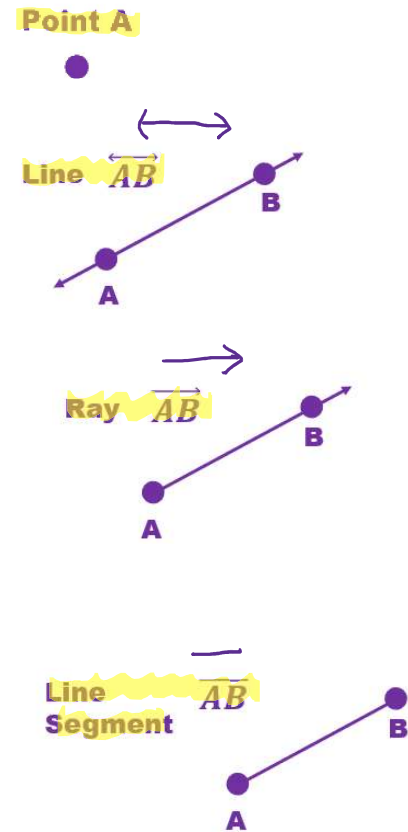


Axioms

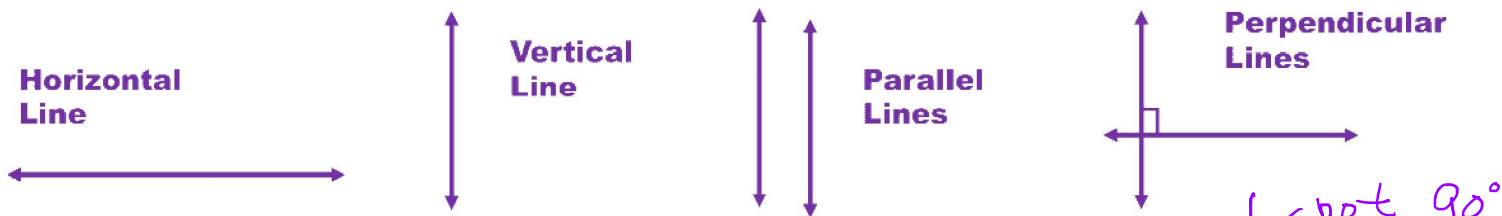
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Some Axioms

- **Point** - that which has no part, no length, width, depth, but just a location in space. In Math, points are defined by a Capital letter (A, B, C, ...) or a Capital Letter with a numeric subscript (P_1 , P_2 , P_3 , etc.).
- **Line** - a straight path of points that continue infinitely in opposite directions. In Math, lines are defined by any two points through which the line passes with a double arrow line above the points ($\overleftrightarrow{AB}$). Sometimes, though unusual, we might see lines defined as a single small letter, such as p, q, r.
- **Ray** - a straight path of points with one terminal point and the other end that continues infinitely. A ray is also defined by two points, the terminal point and any other point, and a single arrow above the points ($\overrightarrow{AB}$).
- **Line Segment** - a straight path of points with two terminal points. A line segment is defined by its terminal points and a line above the points ($\overline{AB}$).



Saturday, May 18, 2024 7:49 AM



Answer the questions about the drawing to the right.

1. How many lines are shown in the drawing?

3: p, q , and r

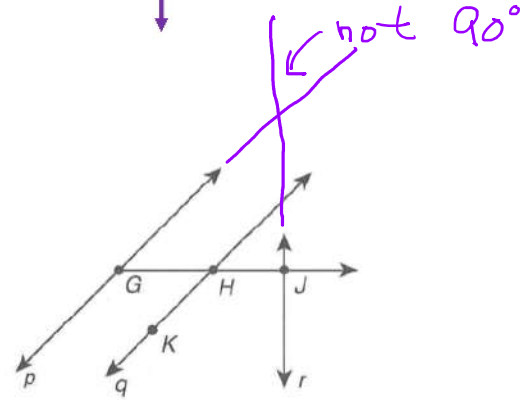
2. Point J lies at the intersection of perpendicular lines. Are lines r and p perpendicular? Explain your reasoning. Line q is BA

No. r & p are NOT perpendicular.

3. Name 4 line segments found in the drawing.

GH, HJ, GJ, KH

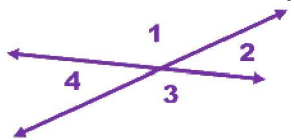
4. There are many rays in the drawing. Name two.

$$\overrightarrow{GJ}, \overrightarrow{HJ}$$


Angles

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Lines that intersect cross at exactly 1 point. The intersection of two lines creates 4 angles.



Vertical Angles are opposite each other $\angle 4$ & $\angle 2$ or $\angle 1$ & $\angle 3$

Acute angle $\angle A < 90^\circ$



Right angle $\angle A = 90^\circ$



Obtuse angle $\angle A > 90^\circ$



Straight angle $\angle A = 180^\circ$ this is a straight line



Complementary angles $\angle A + \angle B = 90^\circ$



Supplementary angles $\angle A + \angle B = 180^\circ$



These are just examples; there are an infinite number of angles

Angles, cont'd...

Monday, May 27, 2024 6:53 PM

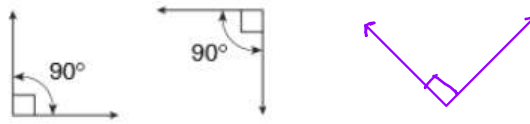
OR (for example)

Name of Angle **Properties**

Examples

right angle

exactly 90°



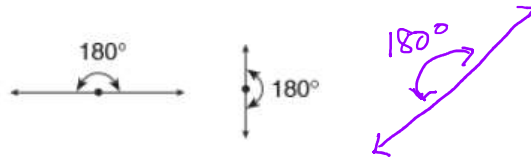
acute angle

less than 90°



straight angle

exactly 180°



obtuse angle

between 90° and 180°



don't worry about this!!

Note: an angle of 225° would be called -135° obtuse.

Answer each question.

5. Match the illustrations with the descriptions.

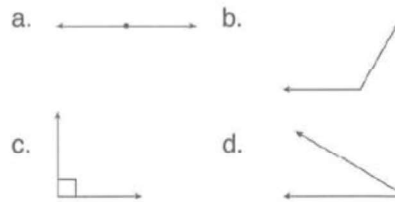
D an acute angle

C a right angle

B an obtuse angle

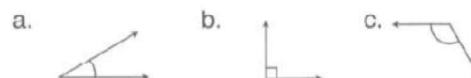
A a straight angle

is this a Right angle?
we do NOT know!!
no $\square$ in the corners!!



6. Which of the three angles is smallest? Why?
(Hint: The size of an angle is based only on the angle, not the length of the sides.)

angle a because it is the only acute angle.



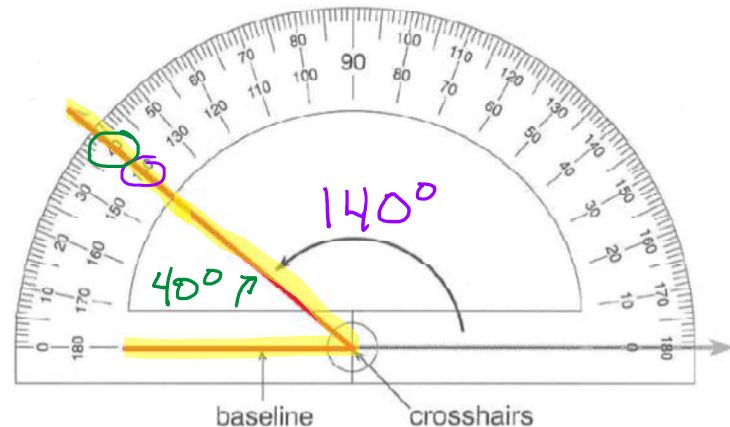
USING PROTRACTORS

A **protractor** is a tool for measuring angles. A protractor looks like a half circle with a fanlike scale of numbers. One scale runs from left to right; the other from right to left.

To measure an angle, line up one ray of the angle with the baseline of the protractor. Put the center point of the protractor (sometimes marked with crosshairs) on the vertex of the angle. Read the scale at the point where the other ray crosses it.

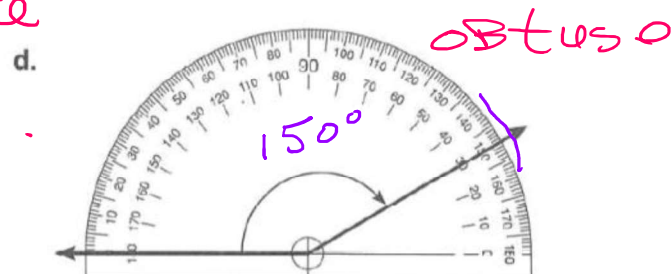
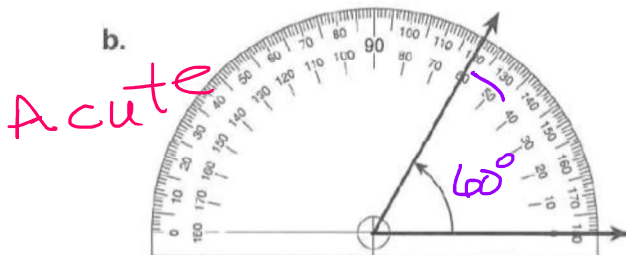
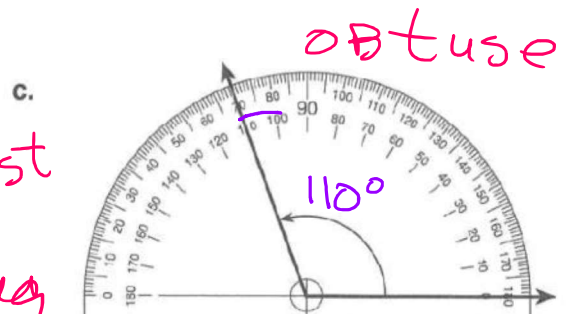
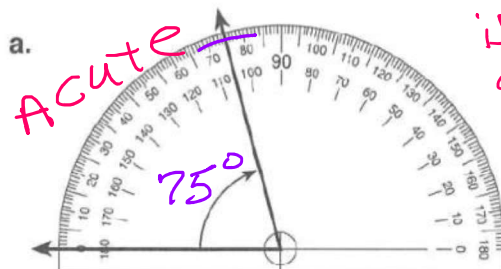
This illustration shows a protractor and an angle of 140° . The symbol $\angle$ means angle. Angles can be named by the vertex point, by a letter or number within the opening, or by three points.

Usually, $\angle A$ or $\angle B$ or ...



Answer each question.

1. Tell the number of degrees in each of the following angles.



if you are not sure, just pick depending on acute or obtuse.

2. Match the angle with the correct number of degrees. Use your judgment or a protractor if you have one.

- b 180°
- a 90°
- c 30°
- d 135°

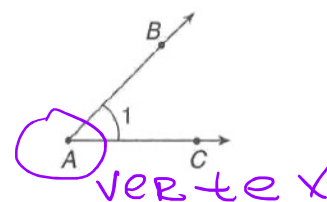


Naming Angels & Arithmetic

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The angle at the right could be named $\angle A$, $\angle 1$, $\angle BAC$, or $\angle CAB$. When three points are used to name an angle, the vertex is always the middle letter.

$\angle A$ OR $\angle BAC$ OR $\angle CAB$
 vertex

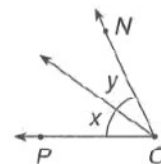


In this illustration, $\angle NOP$ is divided into two smaller angles, $\angle x$ and $\angle y$. If $\angle x = 35^\circ$ and $\angle y = 30^\circ$, we can add to find the measure of $\angle NOP$.

$$35^\circ + 30^\circ = 65^\circ$$

$$\angle NOP = 65^\circ$$

$$m\angle NOP = m\angle x + m\angle y$$



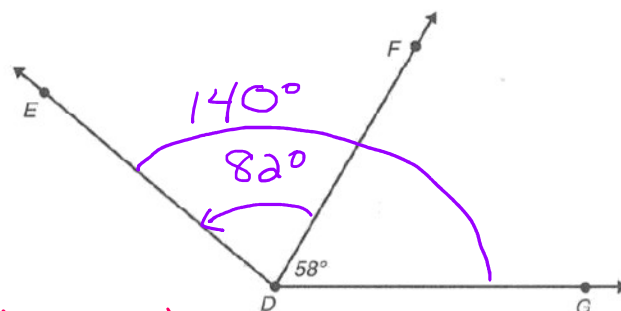
Subtraction can be used to find a missing angle.

(Note: The m before the angle symbol $\angle$ means "the measure of.")

$m\angle EDG = 140^\circ$. Find $m\angle EDF$.

$$m\angle EDG - m\angle FDG = m\angle EDF$$

$$140^\circ - 58^\circ = 82^\circ$$

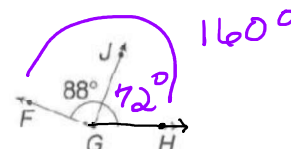


Notice: Everyone forgets to write "m" for "measure of".
 Solve the following problems without using a protractor.

3. $\angle FGH = 160^\circ$. Find the measure of $\angle JGH$.

$$\angle JGH = \angle FGH - \angle FGJ$$

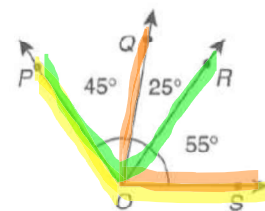
$$= 160^\circ - 88^\circ = 72^\circ$$



4. Tell the type of angle and number of degrees for each angle:

Angle	Type	Degrees
$\angle POS$	Obtuse	$45^\circ + 25^\circ + 55^\circ = 125^\circ$
$\angle QOS$	Acute	$25^\circ + 55^\circ = 80^\circ$
$\angle POR$	Acute	$45^\circ + 25^\circ = 70^\circ$

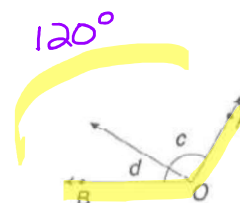
Hint: never trust a picture - determine degrees before identifying the type of angle.



5. $\angle AOB = 120^\circ$ and $\angle c$ is 3 times as great as $\angle d$. Use algebra to find the measurements of both $\angle c$ and $\angle d$. (Hint: Let a variable equal the measurement of $\angle d$.)

$$\angle d + \angle c = 120^\circ$$

$$\angle d + 3\angle d = 120^\circ$$



$$4 \angle D = 120^\circ$$

$$\angle D = 30^\circ \quad \& \quad \angle C = 90^\circ$$

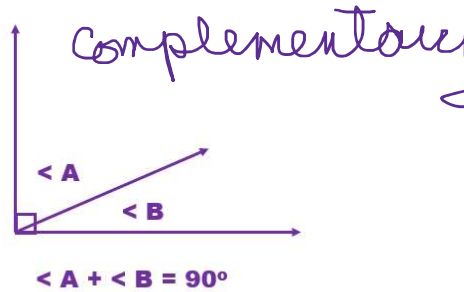
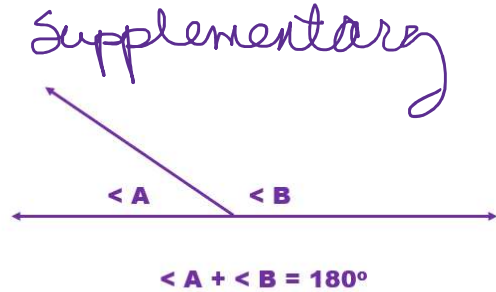
Special Angles

Wednesday, May 29, 2024 7:40 PM

Recall:

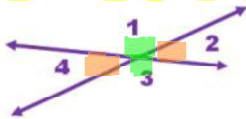
If $\angle A$ and $\angle B$ are Complementary Angles, then their measures add to 90°

If $\angle A$ and $\angle B$ are Supplementary Angles, then their measures add to 180°



the box designates a right (90°) angle

Lines that intersect cross at exactly 1 point. The intersection of two lines creates 4 angles.

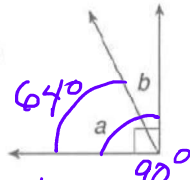


Vertical Angles are opposite each other $\angle 4$ & $\angle 2$ or $\angle 1$ & $\angle 3$

Solve as directed.

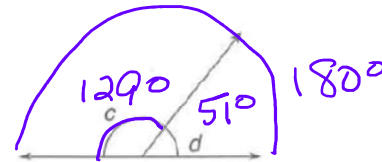
1. If $\angle b = 26^\circ$, what is the measure of $\angle a$?

$$\begin{aligned}\angle a &= 90^\circ - \angle b \\ &= 90^\circ - 26^\circ = 64^\circ\end{aligned}$$



2. If $\angle c = 129^\circ$, what is the measure of $\angle d$?

51°

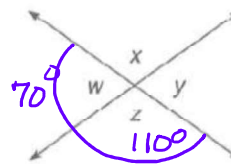


Use the drawing to answer questions 3–5.

3. Name two pairs of vertical angles. w & y ; x & z

4. If $\angle z$ measures 110° , what does $\angle w$ measure? 70°

5. $\angle z = 110^\circ$. What does $\angle x$ measure? How do you know?

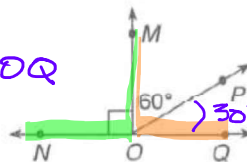


Since $\angle z$ & $\angle x$ are vertical angles, $\angle x = 110^\circ$

Use the drawing to answer questions 6 and 7.

6. Name a supplementary angle to $\angle MON$. $\angle MOQ$

7. What is the measure of $\angle POQ$?



Because

$$\angle MON + \angle MOQ = 180^\circ$$

$\angle MOP$ & $\angle POQ$ are complementary angles, 90° total

also, since $\angle MON = 60^\circ$
 $\angle POQ = 30^\circ$

~~--- perpendicular~~
angles 90° total. $\rightarrow$ since $\angle MON = 90^\circ$
 $\angle POQ = 30^\circ$ Then $\angle MOQ = 90^\circ$